

CAROLLINE Validation Report

Campaign: nominal

Generated: 2026-07-22T16:04:50.732102+00:00

Runs: 5

Success rate: 0.0%

Failure rate: 100.0%

Simulation settings:

duration=120.0s timeout=320.0s

base_seed=42

controller_config=carolline_control/config.yaml

Controller gains (nominal):

k0mega: 1.6

k0mega_pre: 2.0

kR: 5.5

kR_pre: 3.5

kv: 6.0

kx: 4.5

rolling_kd: 1.05

rolling_kp: 0.65

Statistical summaries:

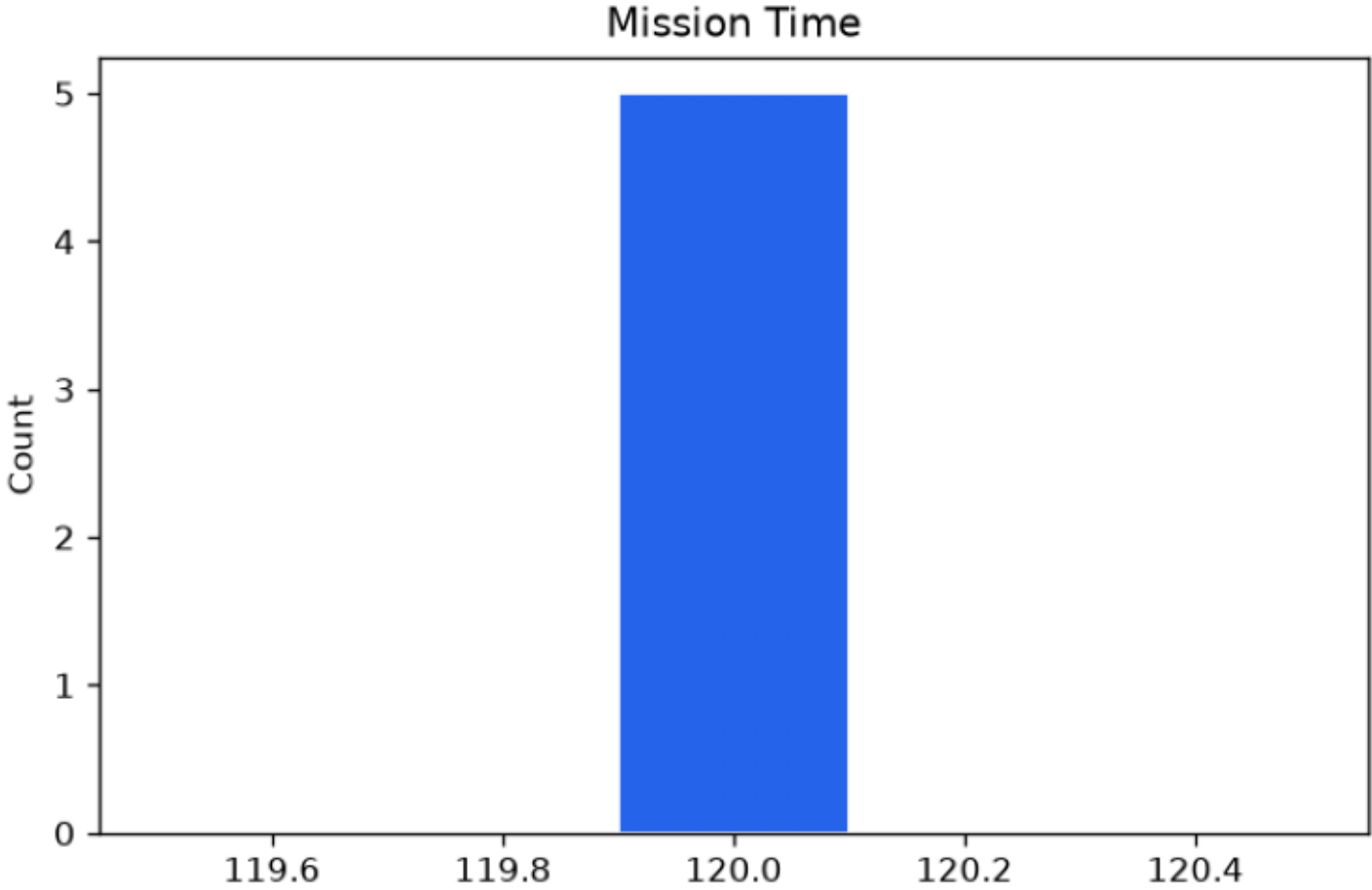
mission_time: mean=120.0000 std=0.0000 95%CI=[120.000, 120.000]

max_position_error: mean=7.4583 std=1.0474 95%CI=[6.540, 8.376]

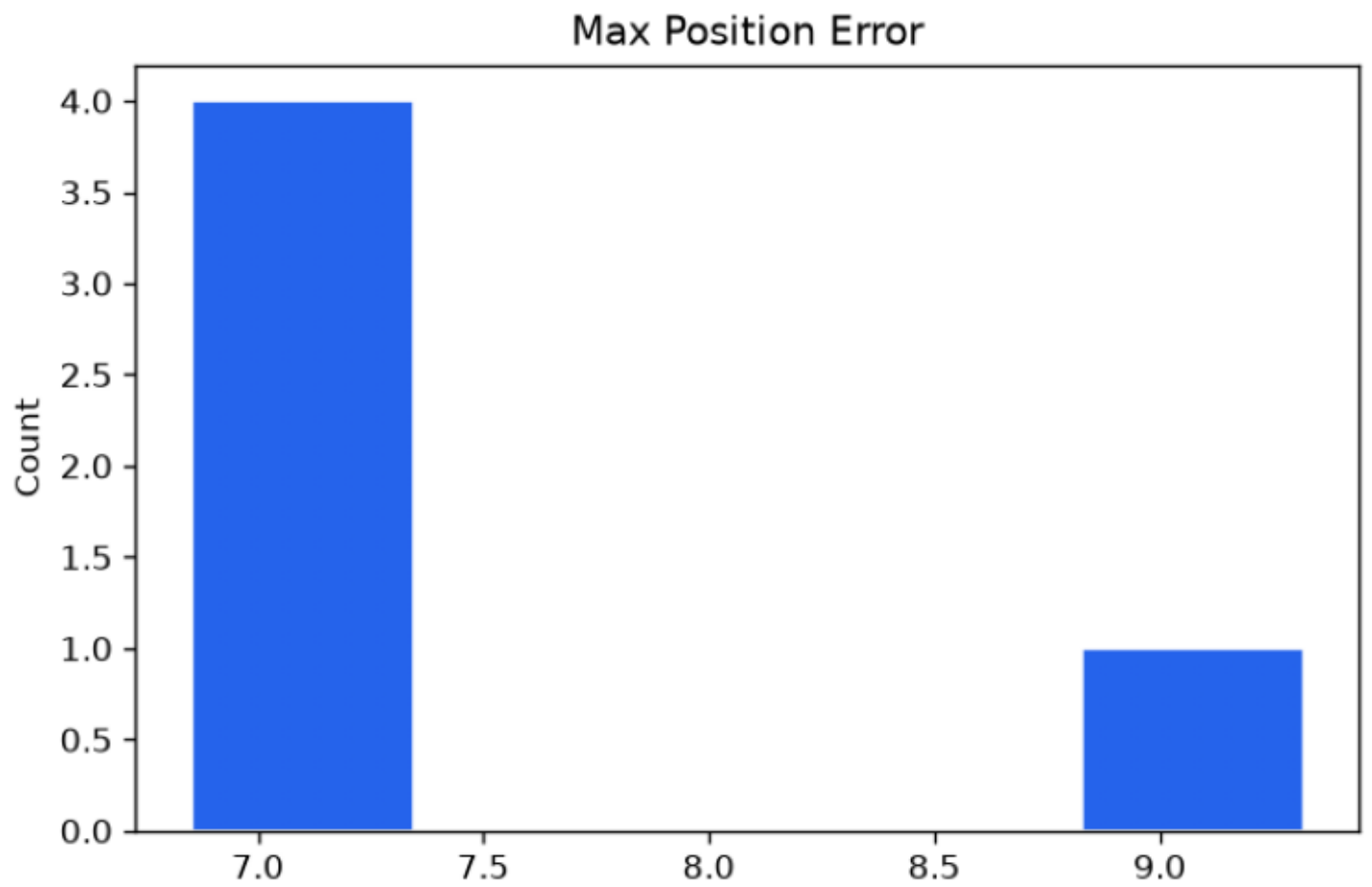
max_tilt_deg: mean=173.4146 std=8.4456 95%CI=[166.012, 180.818]

total_energy_j: mean=2302.1364 std=287.5979 95%CI=[2050.046, 2554.227]

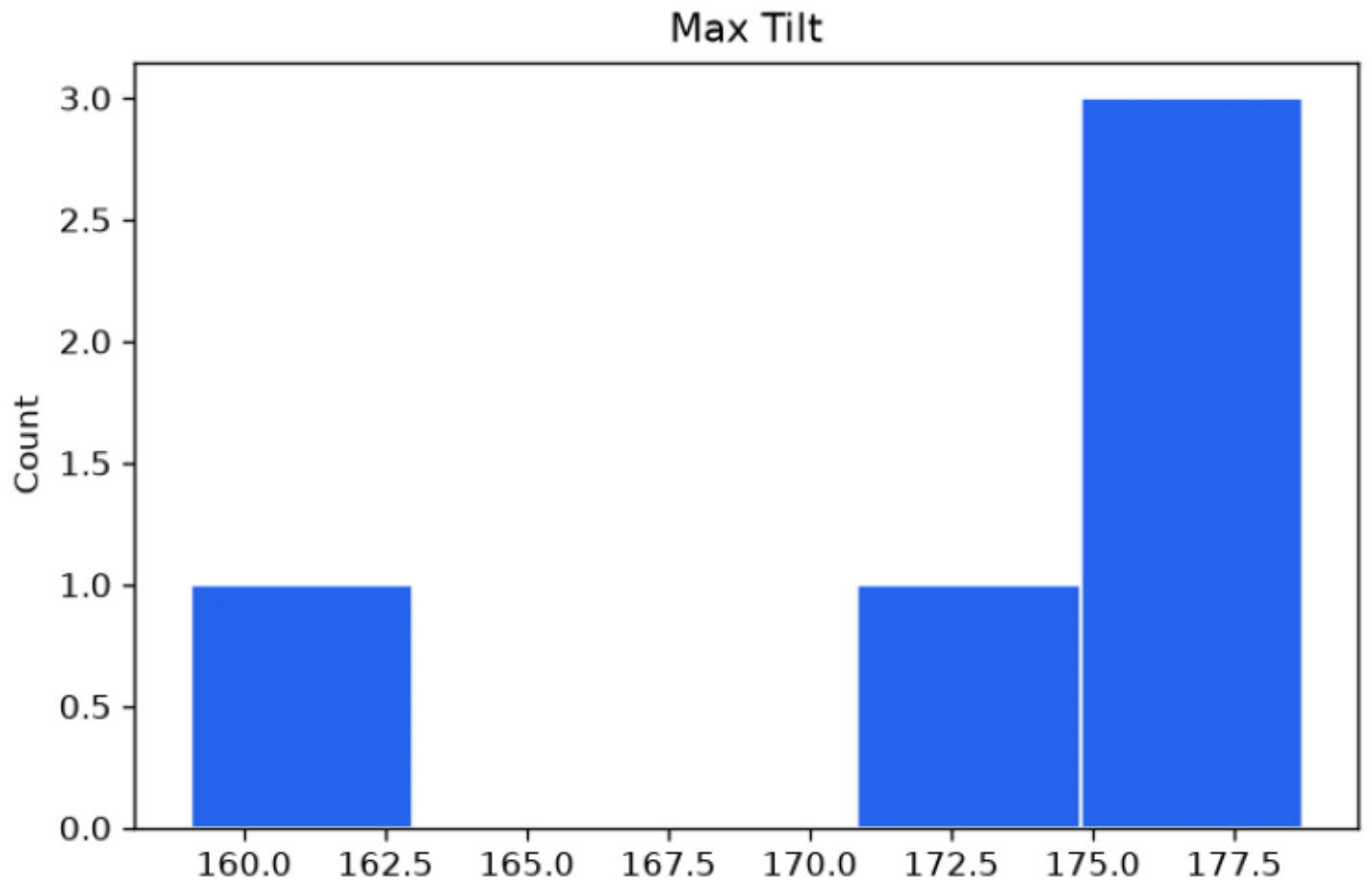
Mission Time Distribution



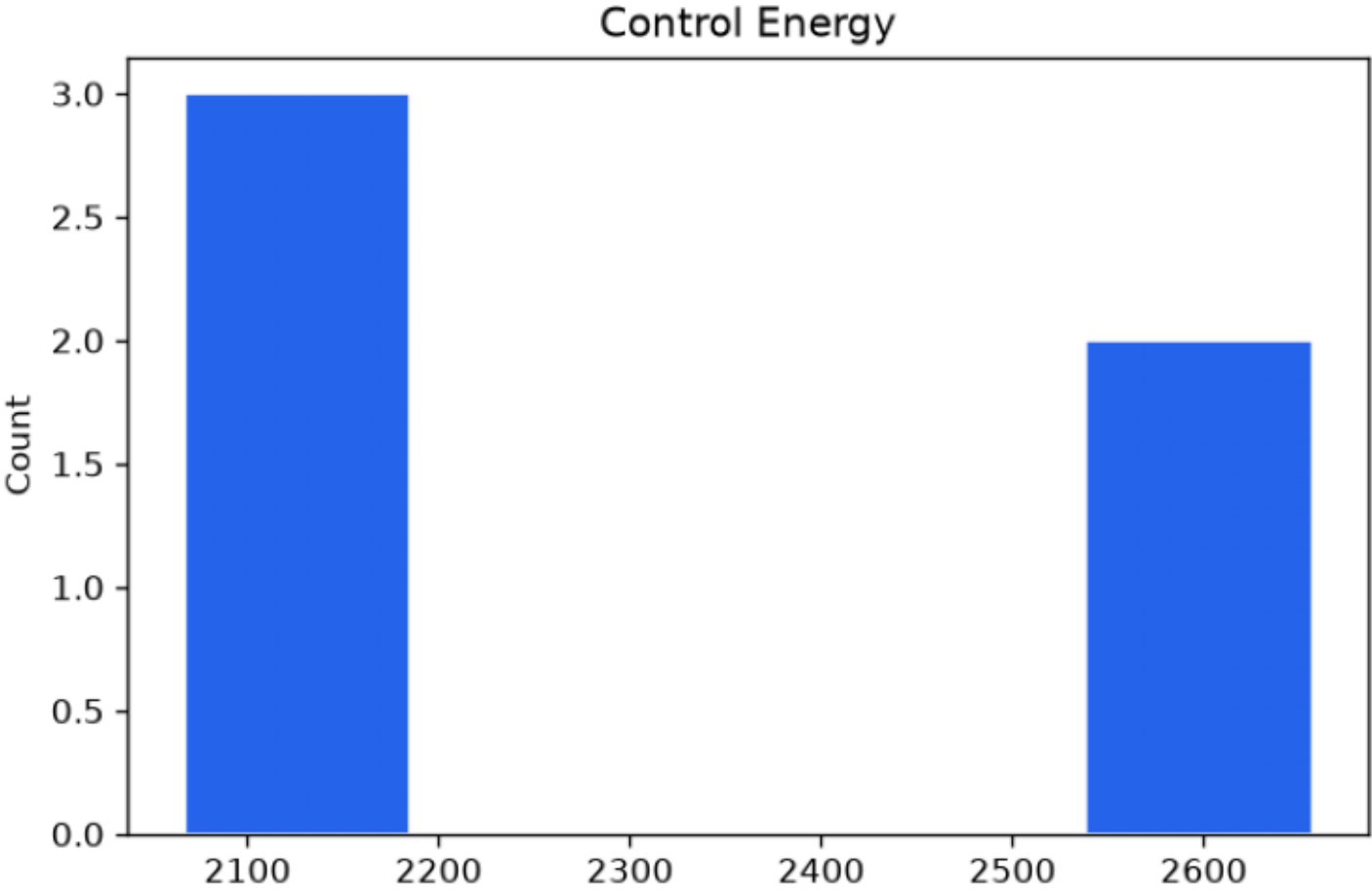
Position Error Distribution



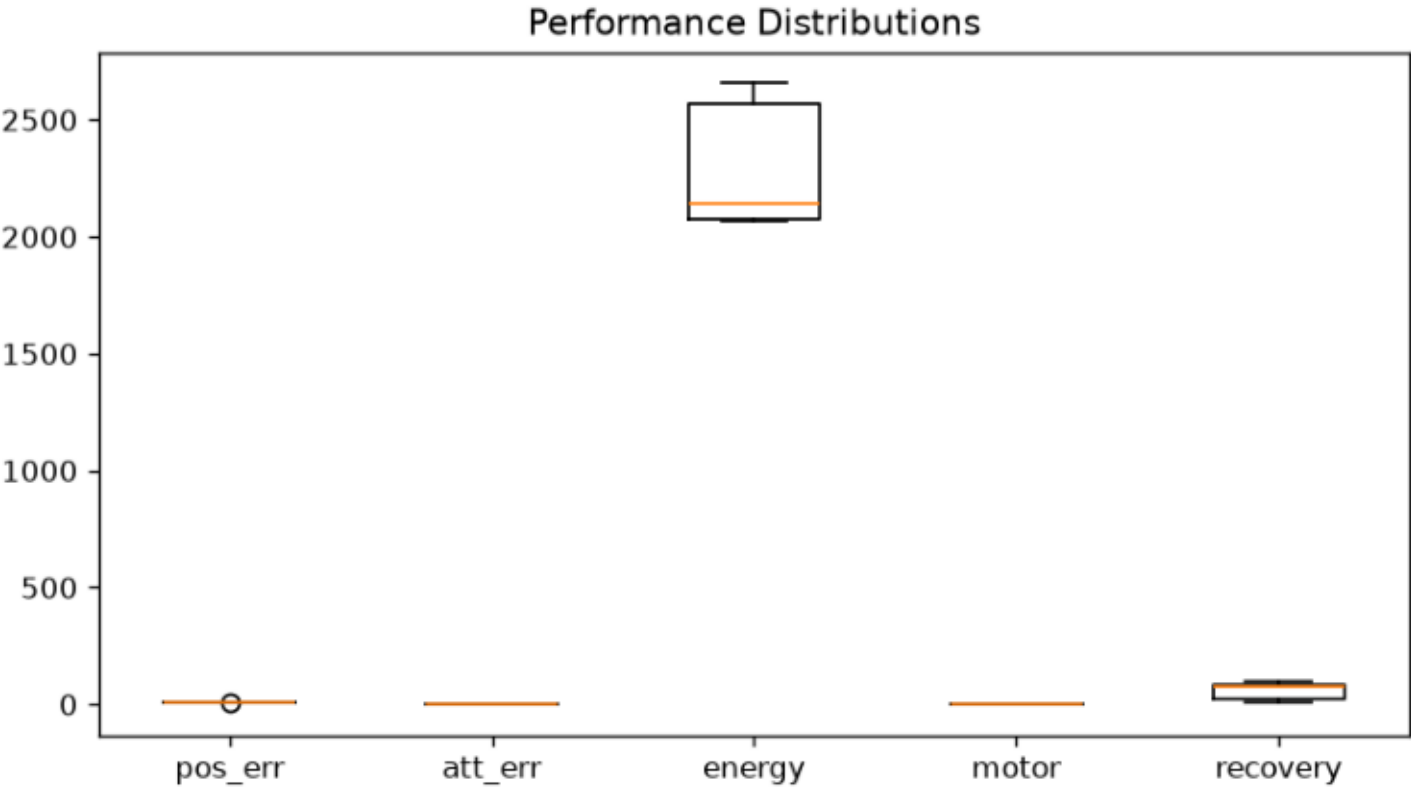
Tilt Distribution



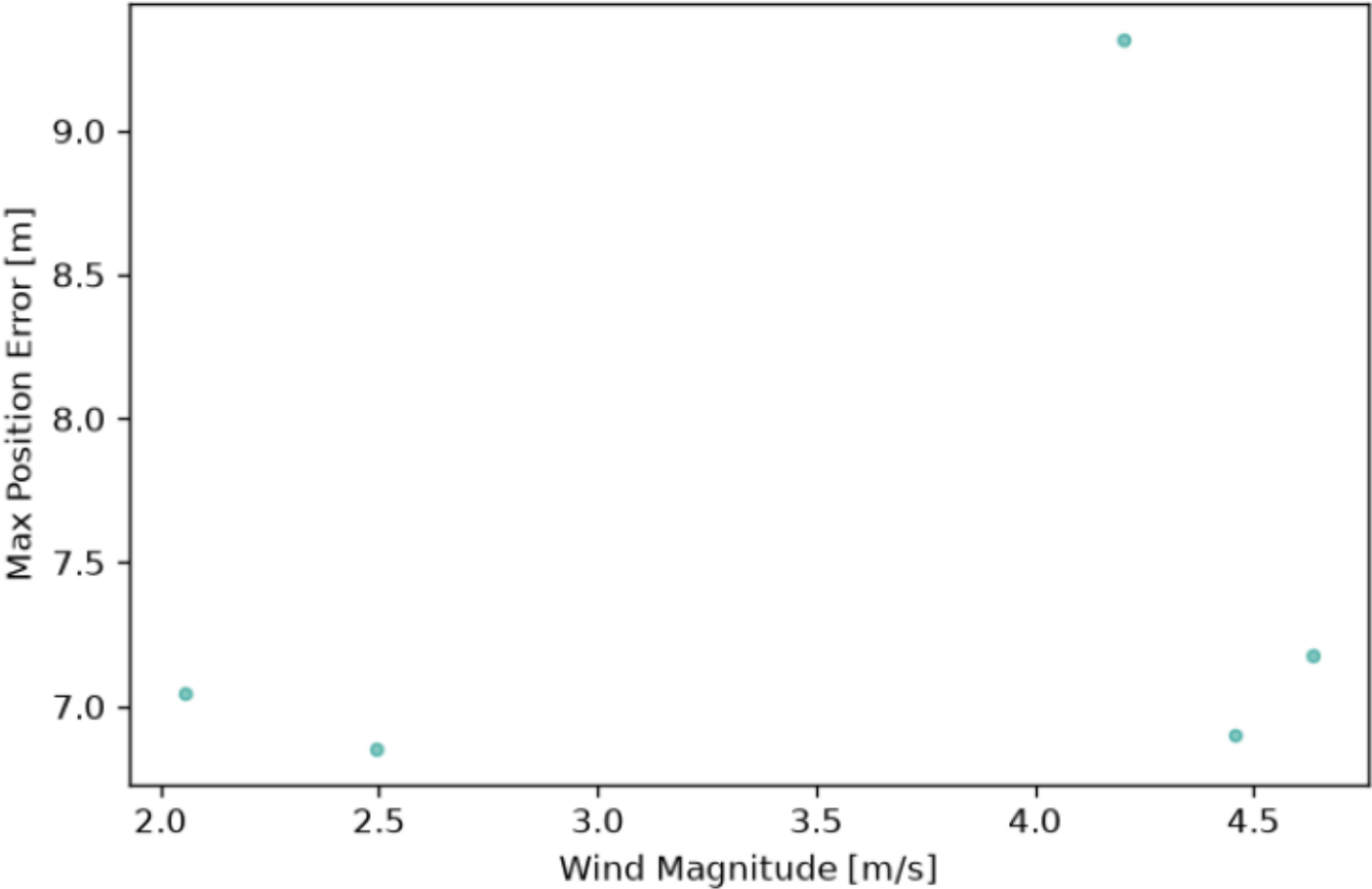
Energy Distribution



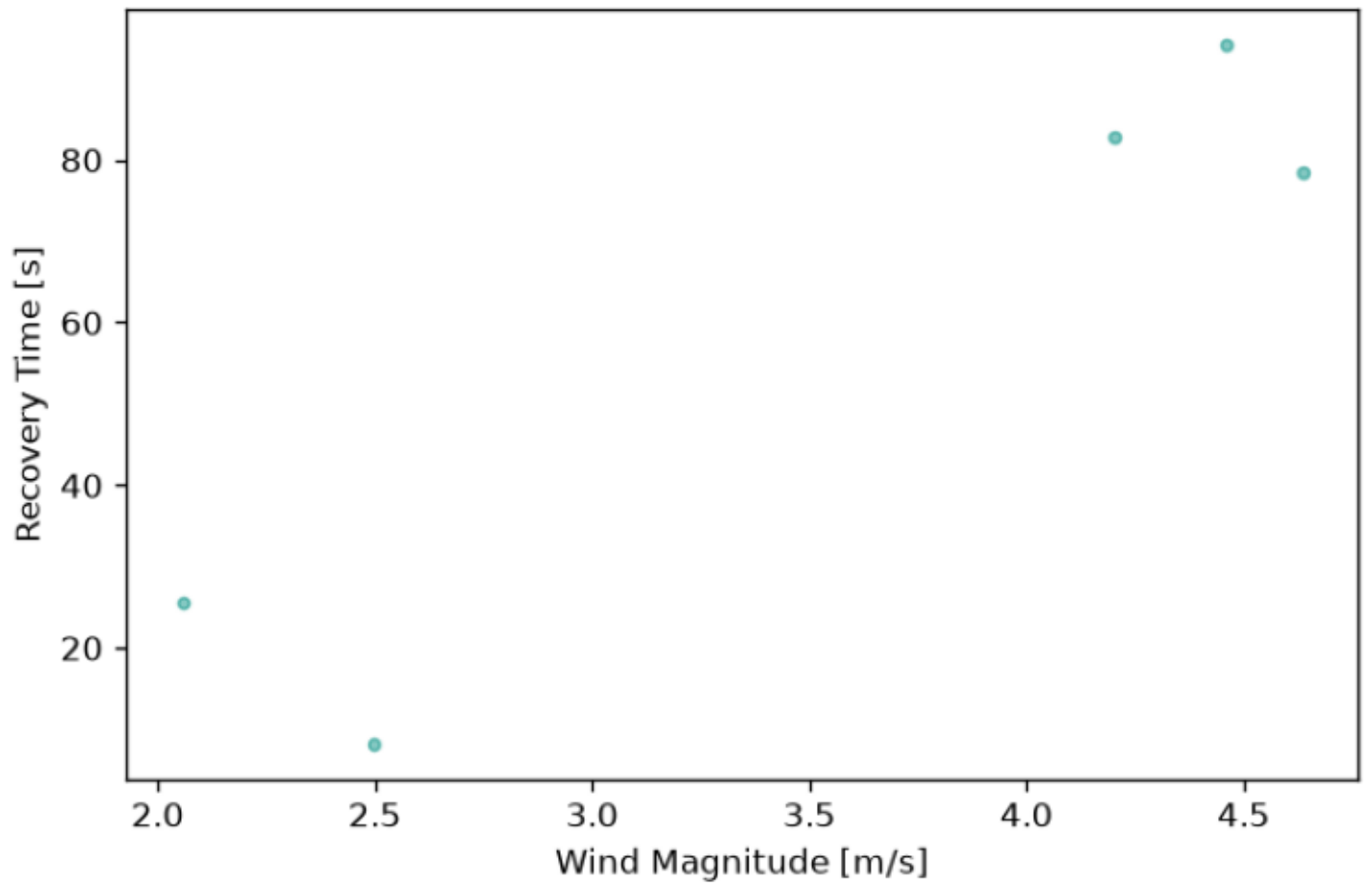
Performance Box Plots



Wind vs Tracking Error



Wind vs Recovery Time



Mass vs Energy

